

SHARDUL HIROLIKAR

+91-8857058550 | shardulhirolikar@gmail.com | github.com/shiro1307 | linkedin.com/in/shardul-hirolikar

SUMMARY

Third-year Computer Science (AI) student at VIT Pune (9.22 GPA) and granted patent holder with deep experience building full-stack AI pipelines and performance-critical systems. Proven problem solver (300+ LeetCode) specializing in scalable full-stack architecture and low-level software engineering.

EDUCATION

Vishwakarma Institute of Technology, Pune | B.Tech, Computer Science (Artificial Intelligence) | 2024 – 2028 | CGPA: 9.22/10

Coursework: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Foundations of AI & ML, Computer Architecture

TECHNICAL SKILLS

Languages: C++, JavaScript, TypeScript, Python, SQL, C, Lua

Frameworks & Libraries: Next.js, React.js, Node.js, Express.js, Flask, Streamlit, NumPy, Pandas

Databases & Tools: Supabase (PostgreSQL), Firebase, MySQL, Git, Linux/Unix Terminal

Core Concepts: Data Structures & Algorithms (LeetCode 300+), Low-Level Memory Management, Cache Eviction Policies, Numerical Simulations, Multi-Tenant Architecture

PROJECTS

AceMyInterview – AI-Powered Mock Interview Platform | Next.js · TypeScript · Supabase · Gemini API

- Shipped a full-stack AI interview platform in a **10-day sprint**, spanning **10+ core features** and complete user lifecycle management.
- Built a *single-pass Gemini evaluation pipeline* consolidating multi-question data into one API call, cutting token overhead **40%+** and reducing latency.
- Designed a normalized **7-table PostgreSQL schema** with Supabase Auth + *Row-Level Security* enforcing strict multi-tenant isolation.
- Integrated native *Speech-to-Text/Text-to-Speech* for a hands-free, low-latency interview experience with state recovery.

EchoDeck – AI-Powered Collaborative Flashcard Platform | Flask · Firebase · Gemini API

- Built an end-to-end Gemini pipeline parsing multi-page PDFs into *structured, relational JSON* for instant flashcard generation.
- Designed a deck sharing/forking system with public/private configs and **granular permissions**.

Systems & Low-Level Modules | C++ · C · Python

- Engineered a custom *LRU cache* (Hash Map + Doubly Linked List) with guaranteed **O(1)** lookup/eviction.
- Built a `std::vector` alternative in C++ with manual heap allocation/resizing, plus a `HashSet` using *linear probing*.
- Implemented memory-efficient *A* pathfinding* in raw C for constrained grid navigation.

Real-Time Cloth Physics Engine | JavaScript · HTML5 Canvas

- Built a *Verlet-integration* engine simulating **3,200+ particles / 7,200+ constraints** at **820+ FPS** (sub-2.5ms frames).
- Eliminated GC overhead via *TypedArrays* (`Float32Array/Uint16Array`); implemented a *4-pass Gauss-Seidel solver* with real-time mesh cutting.

Retro 2.5D Graphics Renderer | JavaScript · C++

- Built a Wolfenstein-3D-style *raycasting engine* computing real-time ray-plane intersections on HTML5 Canvas.
- Developed a dependency-free C++ vector library (*Vector3D-mini*) for matrix ops, coordinate projection, and dot/cross products.

ACHIEVEMENTS

- Patent (Granted):** Co-invented a smart retrofit hardware fan regulator using **Fuzzy Logic control**; architected the fuzzy control logic, simulated thresholds, and validated real-world hardware prototyping.
- Arbitrage Arena 2026 (IISc Bangalore) – National Semi-Finalist:** Developed an **LSTM financial time-series forecasting model** with custom feature engineering to **backtest trading strategies**.